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AN2DL - Second Challenge Report
Overfit & Chill

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1 Introduction

This project focuses on *text classification* of histology's WSI of **Mammary Tissue** divided in four different classes. This task consists in performing *multiclass classification* on the described dataset using deep learning techniques, in particular using **Convolutional Neural Networks**.

2 Problem Analysis

2.1 Dataset Analysis and Pre-processing

In order to gain a better understanding of the problem, the provided dataset has been inspected and analyzed, focusing particularly on Outliers Detection. To perform this task, **Principal Component Analysis (PCA)** has been implemented:

- **PCA** was used to reduce the dimensionality of the dataset to be able to represent it in a tow-dimensional plot, this lead to the identification of two types of outliers:

The process lead to the identification of two types of outliers:

Outlier Type	Quantity
Shrek	60
Slime-stain	50

After the removal of the contaminated images this was the class distribution.

Class Name	Quantity(%)
Luminal A	157(27%)
Luminal B	203(35%)
HER2(+)	151(26%)
Triple Negative	70(12%)

2.2 Image Augmentation

In order to increase the number of samples in the dataset, and since histology WSI are rotation invariant, it has been decided to implement Data-Augmentation, through a set of simple augmentations, as Rotations and Color Jittering: unlike training on natural images, where the rotation may change the context of the visual data, rotating a histopathology patch does not change the context and it improves the learning process for better reliable embedding learning.

2.3 Dataset Splitting

In order to maintain the dataset distribution equal through both the training and validation split, the dataset was split using Stratified Split, with a ratio of **(80%, 20%)**.

3 Method

We propose a *mask-guided multi-crop classification* pipeline based on an EfficientNet backbone and an attention module explicitly conditioned on the available segmentation masks.

3.1 Multi-crop generation from masks

Each histology image is paired with a binary tissue/tumor mask. Rather than uniformly resizing the full image — which may suppress small but discriminative morphological details — we generate multiple informative views by extracting crops driven by the mask:

The binary mask is decomposed into connected components, for each of those we compute its bounding box and enlarge it by a fixed padding margin to retain surrounding context.

Each region crop is resized to a fixed resolution (e.g., 384×384) and treated as an independent training sample, inheriting the label of the original image. If no valid components are found, we fall back to using the full image as a single crop.

This procedure both preserves local high-frequency information better than global downscaling and increases the effective number of training samples, since each image can contribute multiple region-specific crops.

3.2 Backbone and mask-conditioned attention

Each crop is processed by an ImageNet-pretrained EfficientNet-B0. To exploit the mask beyond input cropping, we incorporate a CBAM-like attention block guided by a softened version of the mask. In particular, we apply:

- **Channel attention**, computed via global average and max pooling followed by a lightweight MLP.
- **Spatial attention**, obtained from the channel-refined features by concatenating channel-wise average and max projections.

3.3 Inference: crop aggregation and test-time augmentation

At inference time, the model predicts at crop level. Final image-level predictions are obtained by aggregating crops belonging to the same original image, averaging the corresponding class probabilities. When enabled, test-time augmentation (TTA) is applied and probabilities are averaged across transformations as an additional stabilizing step.

4 Experiments

4.1 Baseline & backbone

We made different experiments using different models and different techniques. We first picked *KimiaNet* since it's pretrained to recognize WSI images like this one, but it behaved very poorly especially for the Triple Negative class. Then, we switched to *ConvNext* decorated with Patching techniques because it should be optimal for tissue micro structures and can capture large context, and after a lot of tries adjusting the Focal Loss value we choose to implement **CutMix & MixUp** to decrease overfitting and lower train and validation loss.

Model Name	**Val F1/Test F1**
KimiaNet	0.31/0.2891
EfifcientNetB0	0.34/0.3685
EfifcientNetB4AttnCutMix	0.34/0.4015
ConvNextAttnCutMix	0.37/0.4028
EfficientNetB0AttnMixUp	**0.38/0.4072**

4.2 Full-image resizing vs mask-driven cropping

The main reason we inspected this is that global resizing to a small input resolution risk losing subtle morphological patterns.
Thereby we tested: single full-image resize training, single crop around mask bounding box, multi-crop extraction from connected components (final). The analysis resulted in multi-crop was the one that improved robustness because it focuses on notable regions, increases the number of training samples and better handles disconnected regions.

4.3 Mask usage: from *crop only* to *attention guidance*

Cropping uses the mask only at input level thus we aimed for other ways to better make use of mask information to influence intermediate representations.

Therefore we immediately came up with various attention mechanisms: from no mask usage beyond cropping to direct spatial attention and finally CBAM + SoftMask guidance.

4.4 Data augmentations

We implemented a unified augmentation pipeline to increase robustness to stain variability and to enlarge the effective training set.
The pipeline consists of: stain normalization via a Macenko normalizer (`torchstain`), paired image–mask geometric and color transformations (*Albumentations*), mask-conditioned generative synthesis using Stable Diffusion inpainting (*Diffusers*).

Despite this, the official augmentation pipeline includes: horizontal/vertical flips, rotations, affine transformations (scale/translation), color jitter, ImageNet normalization.

We noted a more stable and consistent performance with the latter pipeline, we could have refined the former one by dedicating more time to noise handling during train.

4.5 Class imbalance handling

When performing crop extraction, we noted notable differences in the effective label distribution (some classes produce more crops).
Therefore we tested standard random shuffling, class weights in the loss, `WeightedRandomSampler` at crop level (final).

4.6 Optimizer and schedule testing

We were looking for competitive convergence and good generalization at train time when paired with moderate regularization and a freeze-unfreeze schedule.
We started out by picking *AdamW* as the baseline, then we tested out: *Lion* and *Ranger21* with tuned learning rate and weight decay (final), staged unfreezing of the backbone. The analysis confirmed *Ranger21* held the best convergence of the three approaches even without being fine-tuned (as well as with fine-tuning).

4.7 Pseudo-labeling

The main goal of going with pseudo-labeling is to exploit unlabeled test data distribution. The procedure consists in selecting crops whose max class probability exceeds a confidence threshold and retrains briefly.
Tested thresholds: conservative approaches (*thresholds near 0.7*), aggressive approaches (*thresholds near 0.4*). The best outcome were held by aggressive approaches, more specifically almost always an aggressive approach contributed to a $\Delta_{F_1 Score}$ enhancement of $1 - 2\%$.

5 Discussion

The **Crop & Zoom** technique helped us a lot with feature learning and data augmentation. We have been able to limit the overfitting of the model. The model still confused some classes, especially the minority one, although it recognizes very well the mid class (HER2(+), Luminal A).

6 Conclusions

The model we got at the end is just a baseline to reinforce in order to get better results. A future propose is to train more models and use all of them through **ensembling**. Surely with more images useful to train our network to better recognize each label in a better way.

7 Individual Contributions

Each team member took part in almost every aspect of the project, however this was the main work distribution:

- **Ardu Andrea**: New model testing (ConvexNetV2, EfficientB4), Attention implementation, Pseudo Labeling implementation.
- **Cattaneo Lorenzo**: Dataset inspection, Outliers Detection, Hyper-parameter tuning and model refining.
- **Controguerra Simone**: Data Preprocessing, Mask Aware Cropping and EfficientNetB0 Baseline.
- **Furlan Alberto**: Model fine-tuning, augmentation and normalization pipelines.